

Academic Plagiarism Analysis

Plagiarism detection algorithms play an indispensable role in maintaining academic integrity and intellectual honesty.

By employing robust string matching techniques such as the Knuth-Morris-Pratt and Rabin-Karp algorithms, modern systems can efficiently locate exact phrase duplicates across extensive document corpora.

Furthermore, hash tables enable constant-time lookups for token frequencies, which allows precise calculation of vocabulary overlap and sentence-level similarity scores.

Finally, merge sort algorithms provide deterministic $O(N \log N)$ ranking to order multiple candidate submissions based on their computed similarity percentages.